



PROJECT REQUIREMENT DOCUMENT (PRD)

Project Name: Emotion Detector

- **Group Name:** Green
- **Group Member's Names:** A.Vardhini,G.Mounika,K.Ruthika,N.Harshitha
- **Group Captain Name:** N.Lakshmi Harshitha
- **Project ID:** 19

Category: Machine Learning / Computer Vision

Primary Tech Stack: Python, DeepFace, OpenCV, JavaScript (Webcam Access)

One-Sentence Goal: Detect joy, sadness, or anger from a webcam feed.

1. Project Overview

The **Emotion Detector** project is a **real-time emotion recognition system** that uses **Computer Vision** and **Deep Learning** techniques to analyze **human facial expressions**.

The system captures images using a **webcam through JavaScript in Google Colab**, processes the image using **OpenCV**, and performs **emotion analysis using the DeepFace library**.

The project focuses on detecting **three key emotions: joy, sadness, and anger**, and displays the **dominant emotion** directly on the captured image.

This project demonstrates how **AI-based emotion detection** can enhance **Human-Computer Interaction** by making systems more **intuitive and responsive**.

The detected **dominant emotion** is then displayed on the captured image.

2. Objective

The primary objective of this project is to design and implement a **real-time emotion detection system** that can recognize **human emotions from facial expressions** using a

webcam feed. By leveraging **Deep Learning models provided by the DeepFace library**, the system aims to accurately analyze facial features and predict emotional states.

The primary objective of this project is to:

- Develop a **real-time emotion detection system**
- Capture facial images using a **webcam**
- Analyze emotions using the **DeepFace library**
- Classify emotions as **joy, sadness, or anger**
- Display the detected emotion clearly on the screen

3. Requirements

The requirements define the **necessary system capabilities** needed to successfully execute the Emotion Detector project. These requirements ensure that the system functions efficiently, accurately, and in real time while remaining easy to use and implement within the **Google Colab environment**.

3.1 Functional Requirements

The **system** must support **core emotion detection functions**, including **webcam-based image capture, image processing, emotion analysis using the DeepFace model**, and **clear visual display of the detected emotion**.

- The system shall **access the webcam** using JavaScript
- The system shall **capture an image** from the webcam
- The system shall **read and process the image** using OpenCV
- The system shall **analyze facial emotions** using DeepFace
- The system shall **identify the dominant emotion**
- The system shall **display the detected emotion** on the image

3.2 Non-Functional Requirements

The **system** should ensure **smooth performance** and reliability during execution. It must deliver **emotion detection** results with minimal delay while **maintaining accuracy**. The application should also be **user-friendly**, allowing users to easily capture images and view **emotion results** without complexity.

- **Real-time processing**
- **High accuracy in emotion prediction**
- **Ease of use**
- **Efficient execution in Google Colab**

3.3 Software Requirements

The project requires a combination of **programming tools and libraries** that support image processing, deep learning-based emotion analysis, and result visualization. These tools collectively enable seamless integration between webcam input and emotion prediction

- **Python**
- **Google Colab**
- **DeepFace**
- **OpenCV**
- **NumPy, Matplotlib**
- **JavaScript (for webcam capture)**

3.4 Hardware Requirements

- **Laptop or Desktop**
- **Webcam**
- **Minimum 4 GB RAM**
- **Stable Internet Connection**

4. Expected Output

- A **webcam-captured image** displayed on the screen
- The **detected face** shown clearly
- The **dominant emotion (joy, sadness, or anger)** displayed on the image

- Emotion result printed in the **output console**

✓ Final Outcome

The system successfully detects and displays **human emotions in real time** using **facial expressions** through a **Deep Learning-based approach**.

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PROJECT CODE :  Emotion Detector.ipynb

PROJECT PPT : [link](#)